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<u>Theorem</u> Let $|||\cdot|||$ be any submultiplicative matrix norm. Then for every $A \in M_n$,

$$\rho(A) \leq |||A|||$$

$$\left(\rho(A) = \max\{|\lambda| : \lambda \in \sigma(A)\} \right)$$

<u>Proof</u> Let $Ax = \lambda x$, $x \neq 0$, $|\lambda| = \rho(A)$.

Form the matrix

$$X = [x|x|\cdots|x] \in M_n.$$

Then

$$\begin{aligned} AX &= [Ax|Ax|\cdots|Ax] \\ &= [\lambda x|\lambda x|\cdots|\lambda x] \\ &= \lambda X \end{aligned}$$

So

$$\left.\begin{aligned} &|||AX||| = |||\lambda X||| = |\lambda|\,|||X||| \\ \& \quad &|||AX||| \leq |||A|||\,|||X||| \end{aligned}\right\}$$

$$\underset{|||X|||\neq 0}{\Longrightarrow} \quad |||A|||\,|||X||| \geq |\lambda|\,|||X|||$$

$$\underset{X\neq 0}{\Longrightarrow} \quad |||A||| \geq |\lambda| = \rho(A) \quad \blacksquare$$

Back to perturbations

Theorem Let $A \in M_n$ be diagonalizable.

Suppose $A = S \Lambda S^{-1}$, $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \ldots, \lambda_n)$.

Suppose $|||\cdot|||$ is a submultiplicative matrix norm such that

$$|||D||| = \max\{|d_i| : i = 1, 2, \ldots, n\}$$

for every $D = \mathrm{diag}(d_1, d_2, \ldots, d_n)$.

Let $E \in M_n$. If $\hat{\lambda} \in \sigma(A+E)$, then there exists $\lambda_i \in \sigma(A)$ such that

$$|\hat{\lambda} - \lambda_i| \le |||S||| \, |||S^{-1}||| \, |||E|||.$$

Proof

$$S^{-1}(A+E)S = S^{-1}AS + S^{-1}ES$$
$$= \Lambda + S^{-1}ES$$

So $\hat{\lambda} \in \sigma(A+E) = \sigma(\Lambda + S^{-1}ES)$ (similarity).

Thus $\hat{\lambda} I - \Lambda - S^{-1}ES$ is a singular matrix. Two cases ensue:

(1) $\hat{\lambda} I - \Lambda$ is singular

Then the theorem holds trivially ($|\hat{\lambda} - \lambda_i| = 0$ for some λ_i)

(2) $\hat{\lambda} I - \Lambda$ is nonsingular (invertible).

Then ----

$$\cdots (\hat{\lambda} I - \Lambda)^{-1} \Big(\underbrace{(\hat{\lambda} I - \Lambda) - S^{-1} E S}_{\text{singular}} \Big)$$

$$= I - \underbrace{(\hat{\lambda} I - \Lambda)^{-1} S^{-1} E S}$$

which <u>must</u> be singular. This means

$$1 \in \sigma\left((\hat{\lambda} I - \Lambda)^{-1} S^{-1} E S \right).$$

So by previous theorem,

$$|||(\hat{\lambda} I - \Lambda)^{-1} S^{-1} E S||| \geq \rho\left((\hat{\lambda} I - \Lambda)^{-1} S^{-1} E S\right) \geq 1$$

Thus

$$1 \leq |||(\hat{\lambda} I - \Lambda)^{-1} S^{-1} E S||| \leq |||S^{-1} E S||| \cdot |||\underbrace{(\hat{\lambda} I - \Lambda)^{-1}}_{\text{diagonal!}}|||$$

$$\cancel{\leq |||S^{-1} E S||| \cdot |||(\hat{\lambda} I - \Lambda)^{-1}|||}$$

(because $\cancel{e_i := (\hat{\lambda} I - \Lambda)^{-1}}$,

$$= |||S^{-1} E S||| \cdot \max_i |\hat{\lambda} - \lambda_i|^{-1}$$

$$= |||S^{-1} E S||| \frac{1}{\min_i |\hat{\lambda} - \lambda_i|}$$

$(\min |\hat{\lambda} - \lambda_i| \neq 0$ in case (2))

$\rightsquigarrow$

So

$$1 \le |||S^{-1}ES||| \frac{1}{\min_i |\hat{\lambda} - \lambda_i|}$$

$$\Rightarrow \quad \min_i |\hat{\lambda} - \lambda_i| \le |||S^{-1}ES|||$$

$$\le \underline{|||S|||\,|||S^{-1}|||}\,|||E|||$$

<u>Notes</u>

1. What are norms that satisfy the requirements: $|||\cdot|||_\infty$, $|||\cdot|||_1$, $|||\cdot|||_2$ e.g.
(Prove that all induced norms are submultiplicative.)

2. $|||S|||\,|||S^{-1}|||$ is known as the <u>condition number</u>, $\kappa(S)$, of S.

Observe that

$$\kappa(S) = |||S|||\,|||S^{-1}||| \ge |||SS^{-1}|||$$

$$= |||I||| = 1$$

3. If $\kappa(S)$ is "large": S <u>ill-conditioned</u>
"small": S <u>well-conditioned</u>

4. What matrices do you know such that $\kappa(S) = 1$?

5. Research (Google): what matrices are ill-conditioned.

e.g. $H = [h_{ij}]$, $h_{ij} = \frac{1}{i-j}$

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From last theorem:

<u>Corollary</u> Let $A \in M_n$ be normal with $\sigma(A) = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$. Let $E \in M_n$. If $\hat{\lambda} \in \sigma(A+E)$, then there exists $\lambda_i \in \sigma(A)$ such that

$$|\hat{\lambda} - \lambda_i| \le |||E|||_2$$

<u>Proof</u> By previous theorem for $A = U\Lambda U^*$, $U^*U = I$, $\Lambda = \mathrm{diag}(\lambda_1, \lambda_2, \dots, \lambda_n)$, we know

$$\exists \lambda_i : \quad |\hat{\lambda} - \lambda_i| \le |||U|||_2 \, |||U^*|||_2 \, |||E|||_2$$

~~But $|||U|||_2 \cdot |||U^*||| \ge 1$ &~~

~~$|||U|||_2 \, |||U^*|||_2 \ge |||U^*U|||_2 = |||I|||_2 = 1$~~

But $|||U|||_2 = 1 = |||U^*|||_2$ by the fact that

$$\|Ux\|_2 = \|x\|_2 .$$

<u>Comment</u> For non-diagonalizable matrices, no nice bounds are known! Some results for simple (or normal) eigenvalues do exist!

On eigenvalue differentiation.

Lemma Let $A \in M_n$, $\lambda \in \sigma(A)$ be simple. Suppose $Ax = \lambda x$, $y^* A = \lambda y^*$, $x, y \neq 0$. Then $y^* x \neq 0$

Proof By Schur: $\exists$ unitary U s.t.

$$U^* A U = \begin{bmatrix} \lambda & * \\ 0 & \\ \vdots & B \\ 0 & \end{bmatrix}$$

such that $B \in M_{n-1}$. Recall we take

$$U e_1 = \frac{x}{\|x\|_2}$$

$$(U^* A U e_1 = \cdots)$$

Let z be left eigenvector of $U^* A U$ conv. to λ:

$$\underset{\substack{\uparrow \\ \text{left. eigenv.}}}{\bar{\lambda} z} = (U^* A U)^* z = \begin{bmatrix} \lambda & * \\ 0 & \\ \vdots & B \\ 0 & \end{bmatrix}^* z \qquad (1)$$

If $z = \begin{bmatrix} 0 \\ \xi \end{bmatrix}$, $\xi \neq 0$, then by (1):

$$B^* \xi = \bar{\lambda} \xi ,$$

i.e. $\lambda \in \sigma(B) \subseteq \sigma(A)$, contradiction to λ being simple! So first entry of $z \neq 0$. I.e.,

$$z^* e_1 \neq 0$$

But then,

$$\underbrace{(Uz)^*}_{\text{left. eigenv. of } A} \cdot \underbrace{(Ue_1)}_{x} \neq 0 \quad \blacksquare$$

$$z^* e_1 = z^* U^* U e_1 = z^* U^* \frac{x}{\|x\|_2} \neq 0$$

Theorem

Let $A \in M_n$, $\lambda \in \sigma(A)$ be simple, and

$$Ax = \lambda x, \quad y^* A = \lambda y^*, \quad x^* y = 1, \quad \|x\|_2 = 1$$

Let $A(t) = A + tE$, $E \in M_n$ fixed. Then $\lambda(t) \in \sigma(A(t))$, $\lambda(0) = \lambda$, is a differentiable function of t and

$$\boxed{\lambda'(t) = y^*(t) A'(t) x(t),}$$

where $x(t)$ & $y(t)$ are right & left eigenvectors of $A(t)$ corresponding to $\lambda(t)$, with $x(0) = x$, $y(0) = y$.

In particular:

$$\lambda'(t)\Big|_{t=0} = \lambda'(0) = y^* A'(0) x$$

$$= \boxed{y^* E x = \lambda'(0)}$$

& $$\boxed{\frac{\partial \lambda}{\partial a_{ij}} = \bar{y}_i x_j}$$

Proof Next time.

- No class on Friday.
- The final exam w/ instructions will appear as announcement on Friday.
- Questions: on Monday.